

```
In[*]:= << mGRG`Einstein`
```

Chap. 9 General Relativity Examples

9.1 Reissner-Nordstrom Invariants

9.1.1 Computing the Riemann Tensor and Related Objects

```
In[*]:= coSys = {r, θ, ϕ, t};
metric = {{1/(1 - 2m/r + q^2/r^2), 0, 0, 0}, {0, r^2, 0, 0},
           {0, 0, r^2 Sin[θ]^2, 0}, {0, 0, 0, -(1 - 2m/r + q^2/r^2)}};
```

9.1.2 Invariant Definitions

```
In[*]:= SetDimension[4]
```

```
In[*]:= invOne = ScalarCD[]^2
invTwo = SumDum[RicciCD[la, lb] × RicciCD[ua, ub]]
```

```
Out[*]=
```

R^2

```
Out[*]=
```

$$R_{11} R^{11} + R_{12} R^{12} + R_{13} R^{13} + R_{14} R^{14} + R_{21} R^{21} + R_{22} R^{22} + R_{23} R^{23} + R_{24} R^{24} + \\ R_{31} R^{31} + R_{32} R^{32} + R_{33} R^{33} + R_{34} R^{34} + R_{41} R^{41} + R_{42} R^{42} + R_{43} R^{43} + R_{44} R^{44}$$

In[*]:=

```
SumDum[RiemannCD[la, lb, lc, ld] × RiemannCD[ua, ub, uc, ud]]
invThree = % // TindexSort
```

Out[*]=

$$\begin{aligned}
& R_{1111} R^{1111} + R_{1112} R^{1112} + R_{1113} R^{1113} + R_{1114} R^{1114} + R_{1121} R^{1121} + R_{1122} R^{1122} + R_{1123} R^{1123} + \\
& R_{1124} R^{1124} + R_{1131} R^{1131} + R_{1132} R^{1132} + R_{1133} R^{1133} + R_{1134} R^{1134} + R_{1141} R^{1141} + R_{1142} R^{1142} + \\
& R_{1143} R^{1143} + R_{1144} R^{1144} + R_{1211} R^{1211} + R_{1212} R^{1212} + R_{1213} R^{1213} + R_{1214} R^{1214} + R_{1221} R^{1221} + \\
& R_{1222} R^{1222} + R_{1223} R^{1223} + R_{1224} R^{1224} + R_{1231} R^{1231} + R_{1232} R^{1232} + R_{1233} R^{1233} + R_{1234} R^{1234} + \\
& R_{1241} R^{1241} + R_{1242} R^{1242} + R_{1243} R^{1243} + R_{1244} R^{1244} + R_{1311} R^{1311} + R_{1312} R^{1312} + R_{1313} R^{1313} + \\
& R_{1314} R^{1314} + R_{1321} R^{1321} + R_{1322} R^{1322} + R_{1323} R^{1323} + R_{1324} R^{1324} + R_{1331} R^{1331} + R_{1332} R^{1332} + \\
& R_{1333} R^{1333} + R_{1334} R^{1334} + R_{1341} R^{1341} + R_{1342} R^{1342} + R_{1343} R^{1343} + R_{1344} R^{1344} + R_{1411} R^{1411} + \\
& R_{1412} R^{1412} + R_{1413} R^{1413} + R_{1414} R^{1414} + R_{1421} R^{1421} + R_{1422} R^{1422} + R_{1423} R^{1423} + R_{1424} R^{1424} + \\
& R_{1431} R^{1431} + R_{1432} R^{1432} + R_{1433} R^{1433} + R_{1434} R^{1434} + R_{1441} R^{1441} + R_{1442} R^{1442} + R_{1443} R^{1443} + \\
& R_{1444} R^{1444} + R_{2111} R^{2111} + R_{2112} R^{2112} + R_{2113} R^{2113} + R_{2114} R^{2114} + R_{2121} R^{2121} + R_{2122} R^{2122} + \\
& R_{2123} R^{2123} + R_{2124} R^{2124} + R_{2131} R^{2131} + R_{2132} R^{2132} + R_{2133} R^{2133} + R_{2134} R^{2134} + R_{2141} R^{2141} + \\
& R_{2142} R^{2142} + R_{2143} R^{2143} + R_{2144} R^{2144} + R_{2211} R^{2211} + R_{2212} R^{2212} + R_{2213} R^{2213} + R_{2214} R^{2214} + \\
& R_{2221} R^{2221} + R_{2222} R^{2222} + R_{2223} R^{2223} + R_{2224} R^{2224} + R_{2231} R^{2231} + R_{2232} R^{2232} + R_{2233} R^{2233} + \\
& R_{2234} R^{2234} + R_{2241} R^{2241} + R_{2242} R^{2242} + R_{2243} R^{2243} + R_{2244} R^{2244} + R_{2311} R^{2311} + R_{2312} R^{2312} + \\
& R_{2313} R^{2313} + R_{2314} R^{2314} + R_{2321} R^{2321} + R_{2322} R^{2322} + R_{2323} R^{2323} + R_{2324} R^{2324} + R_{2331} R^{2331} + \\
& R_{2332} R^{2332} + R_{2333} R^{2333} + R_{2334} R^{2334} + R_{2341} R^{2341} + R_{2342} R^{2342} + R_{2343} R^{2343} + R_{2344} R^{2344} + \\
& R_{2411} R^{2411} + R_{2412} R^{2412} + R_{2413} R^{2413} + R_{2414} R^{2414} + R_{2421} R^{2421} + R_{2422} R^{2422} + R_{2423} R^{2423} + \\
& R_{2424} R^{2424} + R_{2431} R^{2431} + R_{2432} R^{2432} + R_{2433} R^{2433} + R_{2434} R^{2434} + R_{2441} R^{2441} + R_{2442} R^{2442} + \\
& R_{2443} R^{2443} + R_{2444} R^{2444} + R_{3111} R^{3111} + R_{3112} R^{3112} + R_{3113} R^{3113} + R_{3114} R^{3114} + R_{3121} R^{3121} + \\
& R_{3122} R^{3122} + R_{3123} R^{3123} + R_{3124} R^{3124} + R_{3131} R^{3131} + R_{3132} R^{3132} + R_{3133} R^{3133} + R_{3134} R^{3134} + \\
& R_{3141} R^{3141} + R_{3142} R^{3142} + R_{3143} R^{3143} + R_{3144} R^{3144} + R_{3211} R^{3211} + R_{3212} R^{3212} + R_{3213} R^{3213} + \\
& R_{3214} R^{3214} + R_{3221} R^{3221} + R_{3222} R^{3222} + R_{3223} R^{3223} + R_{3224} R^{3224} + R_{3231} R^{3231} + R_{3232} R^{3232} + \\
& R_{3233} R^{3233} + R_{3234} R^{3234} + R_{3241} R^{3241} + R_{3242} R^{3242} + R_{3243} R^{3243} + R_{3244} R^{3244} + R_{3311} R^{3311} + \\
& R_{3312} R^{3312} + R_{3313} R^{3313} + R_{3314} R^{3314} + R_{3321} R^{3321} + R_{3322} R^{3322} + R_{3323} R^{3323} + R_{3324} R^{3324} + \\
& R_{3331} R^{3331} + R_{3332} R^{3332} + R_{3333} R^{3333} + R_{3334} R^{3334} + R_{3341} R^{3341} + R_{3342} R^{3342} + R_{3343} R^{3343} + \\
& R_{3344} R^{3344} + R_{3411} R^{3411} + R_{3412} R^{3412} + R_{3413} R^{3413} + R_{3414} R^{3414} + R_{3421} R^{3421} + R_{3422} R^{3422} + \\
& R_{3423} R^{3423} + R_{3424} R^{3424} + R_{3431} R^{3431} + R_{3432} R^{3432} + R_{3433} R^{3433} + R_{3434} R^{3434} + R_{3441} R^{3441} + \\
& R_{3442} R^{3442} + R_{3443} R^{3443} + R_{3444} R^{3444} + R_{4111} R^{4111} + R_{4112} R^{4112} + R_{4113} R^{4113} + R_{4114} R^{4114} + \\
& R_{4121} R^{4121} + R_{4122} R^{4122} + R_{4123} R^{4123} + R_{4124} R^{4124} + R_{4131} R^{4131} + R_{4132} R^{4132} + R_{4133} R^{4133} + \\
& R_{4134} R^{4134} + R_{4141} R^{4141} + R_{4142} R^{4142} + R_{4143} R^{4143} + R_{4144} R^{4144} + R_{4211} R^{4211} + R_{4212} R^{4212} + \\
& R_{4213} R^{4213} + R_{4214} R^{4214} + R_{4221} R^{4221} + R_{4222} R^{4222} + R_{4223} R^{4223} + R_{4224} R^{4224} + R_{4231} R^{4231} + \\
& R_{4232} R^{4232} + R_{4233} R^{4233} + R_{4234} R^{4234} + R_{4241} R^{4241} + R_{4242} R^{4242} + R_{4243} R^{4243} + R_{4244} R^{4244} + \\
& R_{4311} R^{4311} + R_{4312} R^{4312} + R_{4313} R^{4313} + R_{4314} R^{4314} + R_{4321} R^{4321} + R_{4322} R^{4322} + R_{4323} R^{4323} + \\
& R_{4324} R^{4324} + R_{4331} R^{4331} + R_{4332} R^{4332} + R_{4333} R^{4333} + R_{4334} R^{4334} + R_{4341} R^{4341} + \\
& R_{4342} R^{4342} + R_{4343} R^{4343} + R_{4344} R^{4344} + R_{4411} R^{4411} + R_{4412} R^{4412} + R_{4413} R^{4413} + \\
& R_{4414} R^{4414} + R_{4421} R^{4421} + R_{4422} R^{4422} + R_{4423} R^{4423} + R_{4424} R^{4424} + R_{4431} R^{4431} + \\
& R_{4432} R^{4432} + R_{4433} R^{4433} + R_{4434} R^{4434} + R_{4441} R^{4441} + R_{4442} R^{4442} + R_{4443} R^{4443} + R_{4444} R^{4444}
\end{aligned}$$

Out[*]=

$$\begin{aligned}
& 4 R_{1212} R^{1212} + 8 R_{1213} R^{1213} + 8 R_{1214} R^{1214} + 8 R_{1223} R^{1223} + 8 R_{1224} R^{1224} + \\
& 8 R_{1234} R^{1234} + 4 R_{1313} R^{1313} + 8 R_{1314} R^{1314} + 8 R_{1323} R^{1323} + 8 R_{1324} R^{1324} + \\
& 8 R_{1334} R^{1334} + 4 R_{1414} R^{1414} + 8 R_{1423} R^{1423} + 8 R_{1424} R^{1424} + 8 R_{1434} R^{1434} + \\
& 4 R_{2323} R^{2323} + 8 R_{2324} R^{2324} + 8 R_{2334} R^{2334} + 4 R_{2424} R^{2424} + 8 R_{2434} R^{2434} + 4 R_{3434} R^{3434}
\end{aligned}$$

```
In[*]:= CD[la, ua, ScalarCD[]]
invFour = SumDum[CDtoBD[CDtoBD[%]]]
```

Out[*]=

$$\nabla_a \nabla^a R$$

Out[*]=

$$\begin{aligned} & \partial_4 \partial^4 R + \partial_3 \partial^3 R + \partial_2 \partial^2 R + \partial_1 \partial^1 R + \partial^4 R \Gamma_{44}^4 + \partial^3 R \Gamma_{43}^4 + \\ & \partial^2 R \Gamma_{42}^4 + \partial^1 R \Gamma_{41}^4 + \partial^4 R \Gamma_{34}^3 + \partial^3 R \Gamma_{33}^3 + \partial^2 R \Gamma_{32}^3 + \partial^1 R \Gamma_{31}^3 + \partial^4 R \Gamma_{24}^2 + \\ & \partial^3 R \Gamma_{23}^2 + \partial^2 R \Gamma_{22}^2 + \partial^1 R \Gamma_{21}^2 + \partial^4 R \Gamma_{14}^1 + \partial^3 R \Gamma_{13}^1 + \partial^2 R \Gamma_{12}^1 + \partial^1 R \Gamma_{11}^1 \end{aligned}$$

```
In[*]:= CD[la, RicciCD[lb, lc]] × CD[ua, RicciCD[ub, uc]]
invFive = SumDum[CDtoBD[%]]
```

Out[*]=

$$\nabla_a R_{bc} \nabla^a R^{bc}$$

Out[*]=

$$\partial_1 R_{11} \partial^1 R^{11} + \partial_1 R_{12} \partial^1 R^{12} + \partial_1 R_{13} \partial^1 R^{13} + \partial_1 R_{14} \partial^1 R^{14} + \partial_1 R_{21} \partial^1 R^{21} + \partial_1 R_{22} \partial^1 R^{22} + \dots 9066 \dots$$

메모리 크기: 2.4 MB

+ 많이 표시

모두 표시

아이콘화 ▼

→ 완전한 수식을 노트북에 저장



9.1.3 Evaluation of the Invariants

```
In[*]:= SetCTensor[coSys, metric, RicciCD → True]
```

```
In[*]:= invOne
```

Out[*]=

$$0$$

```
In[*]:= Metricg[ua, uc] × Metricg[ub, ud] × RicciCD[lc, ld]
ricciUp = Table[SumDum[%], {ua, 4}, {ub, 4}]
```

Out[*]=

$$g^{ac} g^{bd} R_{cd}$$

Out[*]=

$$\left\{ \left\{ -\frac{q^2 (q^2 - 2 m r + r^2)}{r^6}, 0, 0, 0 \right\}, \left\{ 0, \frac{q^2}{r^6}, 0, 0 \right\}, \right. \\ \left. \left\{ 0, 0, \frac{q^2 \text{Csc}[\theta]^2}{r^6}, 0 \right\}, \left\{ 0, 0, 0, \frac{q^4 - 2 m q^2 r + q^2 r^2}{r^2 (q^2 - 2 m r + r^2)^2} \right\} \right\}$$

```
In[*]:= SetComponents[RicciCD[ua, ub], ricciUp]
```

```
In[*]:= invTwo // Simplify
```

Out[*]=

$$\frac{4 q^4}{r^8}$$

```
In[*]:= (Metricg[ua, ue] × Metricg[ub, uf] ×
          Metricg[uc, ug] × Metricg[ud, uh] × RiemannCD[le, lf, lg, lh])
Timing[riemannUp = Table[SumDum[%], {ua, 4}, {ub, 4}, {uc, 4}, {ud, 4}];]
```

Out[*]=

$$g^{ae} g^{bf} g^{cg} g^{dh} R_{efgh}$$

Out[*]=

{0.432592, Null}

```
In[*]:= SetComponents[RiemannCD[ua, ub, uc, ud], riemannUp]
```

```
In[*]:= invThree // Simplify
```

Out[*]=

$$\frac{8 (7 q^4 - 12 m q^2 r + 6 m^2 r^2)}{r^8}$$

```
In[*]:= % /. q → 0
```

Out[*]=

$$\frac{48 m^2}{r^6}$$

```
In[*]:= invFour // Simplify
```

Out[*]=

0

```
In[*]:= Metricg[ua, ud] × GammaCD[ld, lb, uc]
gamma = Table[SumDum[%], {ua, 4}, {lb, -1, -4, -1}, {uc, 4}];
```

Out[*]=

$$\Gamma_{db}^c g^{ad}$$

```
In[*]:= SetComponents[GammaCD[ua, lb, uc], gamma]
```

```
In[*]:= invFive // Simplify
```

Out[*]=

$$\frac{80 q^4 (q^2 + r (-2 m + r))}{r^{12}}$$

CTensor 설정을 해제한다:

```
In[*]:= ClearCTensor[]
```

9.2 Variations and Field Equations

9.2.1 Metric Variations and the Einstein Action

```
In[*]:= Tdefine[Detg[], PrintAs -> "g"]
```

```
In[*]:= lag = Sqrt[-Detg[]] Metricg[ua, ub] x RicciCD[la, lb]
```

```
Out[*]:=
```

$$\sqrt{-g} g^{ab} R_{ab}$$

변분 $\delta\mathcal{L}/\delta g^{ab}$ 계산:

```
In[*]:= VD[Metricg[ua, ub], lag]
% // Absorbg
% // Tsimplify
```

```
Out[*]:=
```

$$\frac{1}{2} \sqrt{-g} R_{ab} + \frac{1}{2} \sqrt{-g} R_{ba} - \left[\frac{g^{cd} R_{cd}}{2 \sqrt{-g}} \left(\delta_{g^{ab}} g \right) \right] + \left[\sqrt{-g} g^{cd} \left(\delta_{g^{ab}} R_{cd} \right) \right]$$

```
Out[*]:=
```

$$\frac{1}{2} \sqrt{-g} R_{ab} + \frac{1}{2} \sqrt{-g} R_{ba} - \left[\frac{R}{2 \sqrt{-g}} \left(\delta_{g^{ab}} g \right) \right] + \left[\sqrt{-g} g^{cd} \left(\delta_{g^{ab}} R_{cd} \right) \right]$$

```
Out[*]:=
```

$$\sqrt{-g} R_{ab} - \left[\frac{R}{2 \sqrt{-g}} \left(\delta_{g^{ab}} g \right) \right] + \left[\sqrt{-g} g^{cd} \left(\delta_{g^{ab}} R_{cd} \right) \right]$$

부분 적분을 통해 $\delta R_{cd}/\delta g^{ab} \rightarrow 0$ 이 된다는 사실을 설정:

```
In[*]:= 1/Sqrt[-Detg[]] VD[Metricg[ua, ub], lag, IndependentVD -> {RicciCD}]
% // Absorbg
einsteinCD = % // Tsimplify
```

```
Out[*]:=
```

$$\frac{R_{ab}}{2} + \frac{R_{ba}}{2} - \frac{\left[\frac{g^{cd} R_{cd}}{2 \sqrt{-g}} \left(\delta_{g^{ab}} g \right) \right]}{\sqrt{-g}}$$

```
Out[*]:=
```

$$\frac{R_{ab}}{2} + \frac{R_{ba}}{2} - \frac{\left[\frac{R}{2 \sqrt{-g}} \left(\delta_{g^{ab}} g \right) \right]}{\sqrt{-g}}$$

```
Out[*]:=
```

$$R_{ab} - \frac{\left[\frac{R}{2 \sqrt{-g}} \left(\delta_{g^{ab}} g \right) \right]}{\sqrt{-g}}$$

부분 적분을 통한 변분 공식 $\delta g / \delta g^{ab} = -g g_{ab}$ 를 설정:

```
In[*]:= VD[Metricg[ua_, ub_], Detg[], rest_, ___] :=  
-rest Detg[] × Metricg[FlipIndex[ua], FlipIndex[ub]] /;  
AllTrue[{ua, ub}, UpIndexQ]
```

```
In[*]:= VD[Metricg[ua, ub], Detg[]]
```

```
Out[*]=  
-g gab
```

결과적으로 아인슈타인 방정식 유도:

```
In[*]:= einsteinCD
```

```
Out[*]=  

$$R_{ab} - \frac{1}{2} g_{ab} R$$

```